

Moderation of Ovulatory Cycle Shifts in Women's Extra-Pair Sexual Interests

by Partner Sexual Attractiveness: Two Preregistered Studies

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Abstract

For decades, researchers have sought to understand how women's sexual psychology shifts across the menstrual cycle, from conceptive status to non-conceptive status, likely under hormonal influence of estradiol and progesterone. One key finding pertains to partnered women: A moderating influence of male partners' sexual attractiveness on the association between conceptive status and women's extra-pair sexual interests (attraction to men other than primary partners). Women partnered with relatively unattractive partners purportedly experience, on average, increased sexual interest in extra-pair men when conceptive in their cycles, compared to non-conceptive phases; women partnered with attractive men do not experience similar increases. A number of studies have found evidence for this association, though recent large-scale studies have yielded mixed and inconclusive evidence. More conclusive research has been needed. In the current two studies, total $N = 484$, naturally cycling women reported their sexual interests daily for 30 days. Analyses yield highly robust partner attractiveness moderation of the association between conceptive status and extra-pair sexual interest. Subsequent analyses support a role for both (imputed via a backward counting method) estradiol and progesterone. These findings have major implications for an understanding of how women's sexual psychology shifts across the cycle.

KEYWORDS: mate choice; sexual selection; mating strategies; estradiol; progesterone; menstrual cycle

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1. Introduction

Women can have sex any day across the ovulatory cycle, but sex can result in conception only during a brief peri-ovulatory phase. To this day, exactly what benefits led women to evolve non-conceptive sex remain unknown, but this much is true: Only when conceptive could women potentially garner genetic benefits for offspring through sex. This observation leads to the speculation that selection shaped women's preferences for certain sexually attractive male features—purported indicators of genetic quality—to be more pronounced when women are conceptive (or, put otherwise, to be *less* pronounced when women are non-conceptive—unable to garner genetic benefits for offspring through sex; e.g., Thornhill & Gangestad, 2008; Gangestad et al., 2023).

This idea, introduced nearly three decades ago (e.g., Gangestad & Thornhill 1998; Thornhill & Gangestad, 1999; Penton-Voak et al., 1999), spawned a large literature examining how women's sexual preferences for male features vary across the cycle. Naturally, it raises the question of what male features may have ancestrally served as cues of male genetic quality. Generally, these features have thought to be ones generally deemed sexually attractive in males (e.g., muscular features; Sell et al., 2017; see also Gangestad & Simpson's, 2000). Traditional meta-analyses and *p*-curve evaluation of this literature produced conflicting results (Gildersleeve et al., 2014a,b; Wood et al., 2014). Furthermore, a wave of large replication studies in the past decade have generally not supported large, robust shifts or have offered mixed (but overall weak) evidence for particular shifts (e.g., on muscularity, see Jünger et al., 2018a; Stern et al., 2021).

Based on ideas about preference shifts, separate predictions were put forward two decades ago. Haselton and her colleagues examined whether shifts in partnered women's extra-pair attraction—attraction to men other than primary partners—depend on whether partners are sexually

attractive. When male partners are not sexually attractive, the prediction goes, women *will* be especially attracted to sexually attractive extra-pair men when conceptive. But not so when partners *are* sexually attractive—the idea being that, when partners offer genetic benefits, it typically does not pay women to look for them elsewhere. (For evidence that, on average, women’s extra-pair partners tend to be more sexually attractive than primary partners, see Murphy et al., 2024.) This is the sort of nuanced, non-obvious prediction that evolutionary thinking can inspire. Multiple studies supported it (Haselton & Gangestad, 2006; Pillsworth & Haselton, 2006; mixed evidence in Larson et al., 2012; Gangestad et al., 2010; Shimoda et al., 2018; see also Gangestad et al., 2005).

In most studies, researchers measured partner sexual attractiveness by asking women to rate their partners’ sexual attractiveness. Hence, these measures reflect women’s perceptions of their partners’ sexual attractiveness, not sexual attractiveness assessed by third parties. Interpretation of these effects must consider this feature. Regardless, moderation effects are of interest.

1.1 Recent examinations of partner attractiveness moderation

Sample sizes in the aforementioned early studies were not impressive (N ranging from 22 to 112). Power to detect the moderating effect of a between-subjects variable (partner sexual attractiveness) on within-woman associations with conceptive status ranged from inadequate to modest (e.g., Blake & Gangestad, 2020). The initial studies were not preregistered as well. Several large-scale studies have more recently examined moderation.

From a large, preregistered daily diary study, Arslan et al. (2021a) claimed that five different ways of assessing partner sexual attractiveness (via women’s perceptions) yielded no evidence for moderation (e.g., partner physical attractiveness, partner short-term attractiveness, partner short-term attractiveness controlling for partner long-term attractiveness). In fact, however, coding errors led four of the five ways to be miscalculated (Gangestad & Dinh, 2021; Arslan et al., 2021b). With errors corrected and analyses re-run, results differed substantially. In analyses using what Arslan et

al. (2021a) described as “best practices” for assessing fertility status (a continuous measure of conception probability, as opposed to discrete fertile “windows,” which yield much less statistical power; Gangestad & Dinh, 2021; see also Schleifenbaum et al., 2024), 4 of 5 ways of examining partner attractiveness moderation of conception probability associations with extra-pair sexual interests were statistically significant (at $\alpha = .05$) in both a preregistered sample of 143 women and a larger (but less “clean”) robustness sample of 421 women (Gangestad & Dinh, 2021). Arslan et al.’s (2021b) reanalyses, which controlled for premenses, found 7 of these 10 moderation effects to be significant (see also Dinh et al., 2022b).¹ Though Gangestad and Dinh (2021) and, in a rejoinder, Arslan et al. (2021b) may have judged the robustness of evidence for moderation differently, the author groups fully agreed on a key point: Additional evidence is needed to assess effects’ size. Arslan et al. (2021b) stated, “the evidence was more mixed and uncertain than our initial brief summary ... made it seem” (p. 445). Gangestad and Dinh (2021) wrote,

[W]e do not claim that Arslan et al.’s data provide definitive evidence for meaningful moderator effects on the relation between fertility and women’s EP sexual interests... In the present instance, point estimates are consistent with substantial and theoretically meaningful moderator effects. At the same time, confidence intervals cannot rule out small true effects (or, naturally, effects much larger than point estimates). Rarely can a single study pinpoint true effect sizes (Maxwell et al., 2015). [p. 440]

Subsequently, Dinh et al. (2022b) examined how women’s extra-pair sexual interests vary across the cycle as a function of changes in ovarian hormone levels. In a sample of 180 naturally cycling women followed across one cycle, partner attractiveness moderated progesterone’s association with a composite measure of extra-pair interests, $\gamma = .09$ (with variables standardized), $p = .003$. When women’s partners were relatively unattractive (1 *SD* below the mean), progesterone levels, as they varied within cycles, negatively covaried with extra-pair interests, $p < .001$. When

¹ In Supplemental Online Materials (SOM), we describe in greater detail results presented in this exchange (see SOM S1a), including analyses using preregistered discrete “fertile windows.”

partners were relatively attractive (1 *SD* above the mean), no association was detected, $p > .25$. As progesterone levels strongly and negatively predict conception probability across the cycle, these results are consistent with Arslan et al.'s corrected findings. Yet they too are less than definitive. The composite measure of extra-pair sexual interest (not explicitly preregistered) consisted of two components. Analyses on separate components yielded robust moderation of progesterone associations with a measure of interest in extra-pair sex, but not extra-pair sexual attraction. Conception probability could be estimated on only a subset of the sample; analyses found no significant moderation of its associations.

In non-preregistered, exploratory analyses on a sample of 209 naturally cycling women, Schleifenbaum et al. (2024) did not detect a conception probability $\times$ partner sexual attractiveness interaction effect on women's extra-pair sexual interests. Yet their measure of women's perceptions of partner sexual attractiveness (a single item asking women to rate the statement, "my partner is very attractive for a one-night stand or short sexual affair," from "does not apply at all" to "applies completely") failed basic validity checks. Past studies (Arslan et al., 2021a; Dinh et al., 2022b) found sizable associations between women's reports of partner sexual attractiveness and their own sexual attraction to partners, in daily reports, with standardized effect sizes averaging roughly .3. (In the current studies, we find the same, very sizable associations.) That's surely as expected: If women view their partners to be sexually attractive, they clearly *should* report sexual attraction to their partners in daily reports, relative to women who view their partners as sexually unattractive. Schleifenbaum et al.'s (2024) item failed this test, standardized effect .03 to .05, non-significant, and roughly five standard errors lower than the mean of prior studies. The distribution of item responses may be telling. On a 5-point scale, 31% of women gave their partner the lowest possible rating, *does not apply at all*. By contrast, in studies straightforwardly assessing partner attractiveness, very few women rate their partners as unattractive (e.g., Arslan et al., 2021a; Dinh et al., 2022b). The unusual

response distribution strongly suggests that women in Schleifenbaum et al.'s study evaluated partners for one-night stands based on features *other than* sexual attractiveness—and, indeed, in support of this interpretation, women's evaluations of partners for one-night stands correlated with these same women's ratings of partners' facial and bodily attractiveness a mere .19 and .17, respectively; even ratings of partners' attractiveness as *long-term* mates predicted these ratings more strongly, $r = .33, .25$. SOM S1b presents detailed analyses pertaining to these issues.

1.2 The current studies

We further assessed the replicability of moderation effects in two preregistered studies. We followed samples of 251 (Study 1) and 233 (Study 2) naturally cycling women across 30 consecutive days. Participants completed a brief survey of their sexual interests each day. Using previously published methods for assessing conception probability (Gangestad et al., 2016, based on Stirnemann et al., 2012; also used by Arslan et al.'s, 2021; Schleifenbaum et al., 2024), we examined moderation by partner sexual attractiveness of associations with a preregistered measure of extra-pair sexual interests. We followed up these analyses with additional analyses examining which ovarian hormone estimates, if either, may moderate associations. Arslan et al. (2023) found that, through counting number of days prior to onset of next menses, one can estimate women's estradiol and progesterone levels, as they change within women's cycles, with validities of .68 to .83, respectively. Though one might worry that these estimates of hormone only allow one to “infer” changes in hormone levels across cycles, rather than directly measure them, in fact that's no different from other measures of hormone levels commonly implemented. What one is typically interested in assessing are the circulating blood levels of a hormone, averaged across a temporal period of interest (e.g., a day). The validity of a measure, then, is its correlation with blood levels. And again, Arslan et al. (2023) found correlations of .68 and .83 for estimates based on backward-counts. They estimated validities of assays of saliva to be poorer: on average, less than .3 for estradiol and roughly .6 for

progesterone. (When LS-MC/MC was utilized, the latter validities averaged .7-.8.) Urinary assays appear to perform better but still no better than estimates from backward counts (e.g., .54 for estradiol, .82 for progesterone; Roos et al., 2015). The evidence indicates, then, that estimates based on backward counts are at least as valid as, and typically *more* valid than, bioassays on saliva or urine.² Methods, measures, and analyses were fully preregistered on Open Science Framework. For preregistrations, data, R-code, and R output files, see [OSE](#). Though additional papers stemming from these data sets are planned (see preregistrations), none have yet been published or are under review.

The two studies differed with respect to some initial questionnaires and daily survey items (reflecting differences in other aims), but procedures testing partner sexual attractiveness moderation were identical. Our Study 2 preregistration included analyses on combined data ($N = 484$), which are our primary analyses. But we also report study-specific analyses and examine moderation by study.

2. Methods

Both studies were reviewed and approved by the [REDACTED FOR ANONYMOUS PEER REVIEW] as having met ethical standards of treatment of human research participants.

2.1 Participants

Participants were recruited through the online platform Prolific. We administered a brief screening questionnaire to select a potential sample of 317/307 (Study 1/Study 2) participants who met criteria: (a) heterosexual romantic relationship of at least 3 months' duration; (b) no hormonal contraceptives (and none in the past 3 months); (c) age 18-35; (d) regular cycles (not varying > 5 days across months); (e) U.S. or U.K resident. All were invited to participate in a study on "Relationship dynamics," which involved an initial online questionnaire, 30 daily surveys, and a

² The validities we report here pertain to assessing variations in hormone level around women's mean level, which is what we seek to do in the current studies. Clearly, one cannot measure differences in mean hormone levels across women using backward counts. In SOM S1c, we present additional details.

follow-up survey. 294/283 participants enrolled. After removal of participants not meeting requirements of minimum level of participation or inclusion criteria (assessed once again in the follow-up), final samples retained for analysis were 251/233 women. On average, they completed 26.5 ($SD = 5.0$) and 24.9 ($SD = 5.7$) daily surveys, respectively (12,475 total daily surveys).

We detail reasons for participant exclusion: In Study 2, several reported in the initial questionnaire that they used hormonal contraceptives (4), were pregnant (1), or had recently experienced a blighted ovum (1), leaving 294 (Study 1)/277 (Study 2) women. 280/252 completed at least 10 daily surveys, a preregistered inclusion criterion. A follow-up questionnaire asked women to report the first day of menses that followed the daily surveys. As well, it asked women to confirm their relationship status and hormonal contraceptive use. An additional 29/19 women were excluded. All but 2/1 failed to meet explicit inclusion criteria: 10/8 did not meet requirements at follow-up (7/3 used hormonal contraceptive in the past month, 0/2 were not in relationship, 1/1 was in relationship with a woman, 2/1 was pregnant, 0/1 reported polycystic ovaries); 13/6 did not have 10+ surveys after short (<18 days) or long (>40 days) cycles were excluded; 4/4 did not have 10+ backward-counted days (a preregistered inclusion criterion) due to missing or insufficient follow-up. In addition, 2/0 did not complete required information about partners on the initial questionnaire permitting examination of moderation; 0/1 had exactly 10 backward-counted days, but no meaningful variation in contraceptive status across these 10 (consecutive) days (conception probabilities ranging from .01-.05, where .55 is maximum). After removal of participants not meeting requirements, final samples retained for analysis were 251/233 women.

In Study 1, one woman reported herself to be 36 years old on the initial questionnaire, despite prescreening selecting her as 35. Possibly, in the week that lapsed between prescreening and the initial questionnaire, she turned 36. We did not exclude her from our analyzed sample.

The validities for backward count estimates of hormone levels were estimated from a sample of 259 women (509 cycles) who participated in the BioCycle study (e.g., Wactawski-Wende et al., 2009). Naturally, if the BioCycle sample had cycles more regular and hormonally normative than the cycles in our sample, the validities Arslan et al. (2023) reported may overestimate validities in the current studies. The BioCycle study screened women with uterine fibroids, current sexually transmitted disease, histories of infertility treatment and endometriosis, various other diseases, and BMI < 18 or > 35. We did not explicitly do so. But we screened women for having regular cycles more strictly than the BioCycle study and eliminated cycles fewer than 18 days or greater than 40 days in length, criteria that should have removed women with highly disruptive uterine conditions such as endometriosis or extreme BMI. The mean cycle length in our studies was 28.9 days, nearly identical to the mean of 29.0 in the BioCycle study. The SD of cycle length was 3.7, less than the SD in the BioCycle study, 4.1 days. Mean age in the two samples was comparable, 29.9 vs. 27.3, but less variable in our sample (SD = 3.8 vs. 8.2). Whereas 84% of our participants cohabited with partners, 26% of BioCycle participants were married or living as married; hence, 94% of our participants reported sex in the month of daily diaries but just 71% of BioCycle participants were sexually active. Women who have regular sex have more regular cycles (e.g., Burleson et al., 1991; Cutler et al., 1980). All in all, the current samples likely display cycles at least roughly as hormonally regular as those in the BioCycle study, if not more so. (We also, however, ran robustness analyses selecting the subsample meeting the BioCycle's criterion for BMI; see *Robustness analyses* below.)

Women were compensated for participation: In Study 1, they received \$1 for the screening questionnaire, \$10 for an initial 30-minute questionnaire, \$2 for each of up to 30 5-minute daily surveys completed, and \$20 for a final 2-minute follow-up survey, for total potential earnings of \$91. In Study 1, they received \$.50 for the screening questionnaire, \$5 for an initial 15-minute

questionnaire, \$1.25 for each of up to 30 5-minute daily surveys completed, and \$10 for a final 2-minute follow-up survey, for total potential earnings of \$53.

2.2 Procedures and measures

Outline. Following the initial screening, all eligible women were sent an invitation to complete an initial 30-minute/15-minute questionnaire for Studies 1 and 2, respectively. All women who did so were then invited to fill out a brief daily survey for 30 consecutive days. For all participants within a study, surveys started and ended on the same calendar days. Afterward, we sent out invitations to a follow-up survey, to be completed only after women had commenced their next menses (with one set of exceptions described below).

Initial questionnaire. The initial questionnaire consisted of several parts. For a full list, see SOM S1d. We asked women about their menstrual cycles. The key preregistered measure pertinent to testing the moderation effects of interest was a measure of women's perceptions of partner sexual attractiveness. Women were asked to respond to the following items on a relative attractiveness scale (1 = *much less*, 7 = *much more*): "How sexually attractive is your partner?"; "How attractive is your partner as a sexual partner?"; "How interested are others in your partner as a sexual partner?" Item responses were averaged to create a composite measure of Partner Sexual Attractiveness ($\alpha = .88/.88$, $\omega_t = .89/.88$, and $\omega = .98/.99$ for Studies 1/2; Revelle & Condon, 2025).

We also asked women to rate themselves on the same items. We averaged responses to create a composite measure of Self Sexual Attractiveness ($\alpha = .84/.90$, $\omega_t = .84/.90$, and $\omega = 1.00/.99$ for Studies 1/2).

In Study 1, the initial survey included three attention-check items, interspersed among survey items: "I have never used a computer"; "Which word is most closely related to 'moody'" (correct response = 'emotional'); "I would be very surprised to see someone eating a sandwich at noon." The last check did not work well, as some women could have been in circumstances in which they would

not observe someone eating a sandwich for lunch. Of the 251 participants who met other criteria, one and six participants “failed” the other two checks, respectively. None “failed” both and, of those who “failed” one, six of seven responded “1” (‘not at all surprised’) to the sandwich item, the remaining participant a “3” on a 7-point scale. As poor responses can arise from inadvertent key strikes as well as lapses in attention, poor response rates were low, and no woman “failed” multiple items, we conservatively did not remove any participant for inattention and did not include the items in Study 2. See SOM S1d, for additional details.

Daily survey. The daily survey consisted of 35/56 items for Studies 1 and 2, respectively. For both studies, the initial items pertained to women’s sexual interests, three of which tapped extra-pair sexual interest. Responses to two items were averaged to create a measure of Extra-Pair Sexual Attraction: “I felt strong sexual attraction toward someone other than my current partner”; “I was strongly physically attracted to someone other than my current partner.” One item measured Interest in Extra-Pair Sex. Women were presented with the following scenario to measure Interest in Extra-Pair Sex: “Imagine that you met a very interesting and attractive person today, and you were very sexually attracted to this person and this person was very sexually attracted to you. Imagine that you have the evening free and you are able to spend the evening with this person.” They were then asked, “How likely do you think you would be to have sex with this person today, if you could be sure that no one would ever find out about it?” (1 = *Not at all likely*, 7 = *Very likely*). To compute a preregistered measure of Extra-Pair Sexual Interest (following Dinh et al., 2022b), we averaged z-scored Extra-Pair Sexual Attraction and Interest in Extra-Pair Sex. Primary analyses used this measure. Follow-up analyses examined these outcomes separately.

Preregistered predictions pertained to extra-pair sexual interests, but we also examined effects on in-pair sexual interest. Responses to two items were averaged to create a measure of In-Pair Sexual Attraction: “I felt a strong sexual attraction toward my primary current partner”; “I was

strongly physically attracted to my primary partner.” One item measured Initiation of Sex in the past day (24 hours): “On how many occasions did you initiate sexual activity with your partner?” To compute a preregistered measure of In-Pair Sexual Interest (following Dinh et al., 2022b), we averaged z -scored In-Pair Sexual Attraction and Initiation of Sex.

Each daily survey asked women to report whether they were currently menstruating and, if so, the calendar day on which current menses began.

Additional items pertaining to sexual interests were General Sexual Desire, Frequency of Sex, Partner-Initiation of Sex, and Rejection of Partner-Initiated Sex. See SOM S1d for item content. For complete lists of items on the daily surveys, see [OSF](#).

Qualtrics offers checks on response quality. Across all 12,475 daily surveys, 0 speeders and 0 low completion rates were detected. Qualtrics technology (available only for Study 2) flagged 6 potential bot respondents (which could be false positives)—a rate of .1%.

Daily survey reliabilities. Participants were assessed across 30 consecutive days (and, within-study, precisely the same days). In SOM S1d, we list R_c and R_{cn} values. Reliabilities were generally good, though R_c for composite Extra-Pair Sexual Interest was modest. As this measure was preregistered, we report primary analyses on it. But we also report on sub-components that are highly reliably. Conclusions do not substantively change.

Follow-up survey. We sent a follow-up survey to 280 and 252 women who completed at least 10 daily surveys in the two samples, per our preregistration. Invitations to complete follow-up surveys were sent out in four waves. The first wave consisted of women whose last session fell on a day on which they were menstruating, in addition to women who never reported a day of menses during the month of daily surveys. These invitations were sent out three days after the last daily survey. The second, third, and fourth waves were sent out one, two, and three weeks after the last daily survey, respectively. Each wave of participants was based on when, within the daily surveys,

women experienced onset of menses: Wave 2: last onset day 1-8; Wave 3: last onset day 9-15; Wave 4: last onset day 16+. We sought to time invitations such that they arrived shortly after or before women experienced onset of next menses.

Follow-up surveys asked women to report day of onset of menses following the daily surveys. We also included items confirming that participants met inclusion criteria: in a heterosexual romantic relationship throughout the entire 30-day period of daily surveys, no use of hormonal contraceptives throughout the entire 30 days, not pregnant, not breastfeeding, female at birth.

Estimates of Conception Probability. Based on reports made on daily surveys and follow-up surveys, backward counts were made from onset of next menses. (-1 = 1 day prior to onset of next menses, -2 = 2 days prior to onset of next menses, etc.) We used these values to estimate day-specific conception probabilities (probability of conception, given unprotected intercourse); see Gangestad et al. (2016), based on Stirnemann et al. (2012). We used the average of a forward count (days from onset of last menses, where day of onset = 1) and a backward count based on typical cycle length to estimate values when we lacked backward counts (see Gangestad et al., 2016); see SOM S1d for details. In total, 192 (2.9%) and 252 (4.3%) of the 12,475 participant-days had estimates through forward counts in Studies 1 and 2, respectively. (By comparison, these percentages were 25.2% and 34.1% in Arslan et al.'s [2021] preregistered and robustness samples, respectively.) Of these, just 77 (0.6%) were non-menstrual days. Our preregistration required at least 10 days of backward counts for a participant to be included. Of the 484 participants, 120 (25%) had at least one day's values estimated through forward counts (range 1-11; $M = 3.8$). Most were ones who experienced menses at the end of the daily survey days, for whom we did not ask for onset of next menses in the follow-up survey; forward count imputed conception probabilities, in these cases, were for menstrual days, near-zero, and accurate. Just 40 participants (8%) had values for at least one non-menstrual day imputed through forward counts (range 1-6; $M = 1.9$).

Estimates of Estradiol and Progesterone levels. Arslan et al. (2023) used archival data on a sample on which serum levels of estradiol and progesterone levels were available (the BioCycle study; Wactawski-Wende et al., 2009) to estimate estradiol and progesterone levels based on backward or forward/backward-typical cycle counts. For estradiol levels, these values have estimated validities of $\sim .68/.57$ (for backward and forward counts, respectively). For progesterone levels, validities are $\sim .83/.72$ (Arslan et al., 2023).

2.3 Analyses

Primary analyses examining conception probability and interactions. We performed preregistered multilevel regression analyses on daily surveys, nested within participant, with Extra-Pair Sexual Interest as the outcome variable and fixed effects of Menses (0 = no, 1 = yes), within-woman Conception Probability, Partner Sexual Attractiveness, and the within-woman Conception Probability $\times$ Partner Sexual Attractiveness interaction. We furthermore controlled for Self Sexual Attractiveness and its interaction with Conception Probability. (Partner and Self Sexual Attractiveness positively covaried and Self Sexual Attractiveness could interact with Conception Probability to affect Extra-Pair Sexual Interest.) Within-woman Conception Probability, Partner Sexual Attractiveness, and Self Sexual Attractiveness were $\bar{x}$ -scored prior to analysis. An interaction with Study was included for each effect. In preregistered analyses, we included as a predictor between-woman Conception Probability ($\bar{x}$ -scored), along with its interactions with Partner and Self Sexual Attractiveness. We also included Menses interactions with study, Partner Sexual Attractiveness, and Self Sexual Attractiveness as control variables if $p < .10$.³

We modeled random intercept variation across participants and used Matuschek et al.'s (2017) iterative procedures to model random slope variation, across participants, for within-woman

³ Study was $\bar{x}$ -scored in these analyses. With interactions with Study included, then, effects for all other terms, including the critical Conception Probability $\times$ Partner Sexual Attractiveness interaction, were weighted by sample size across studies. As we report, Study interactions occurred at chance levels in any case.

Conception Probability. These procedures retain random slope terms (here, for Conception Probability as well as its covariance with random intercepts effects) when necessary for α -control for testing fixed effects, while not compromising power when random slope terms are not needed. See SOM S1d for details. For all models, a random slope term for Conception Probability was retained.

SOM S1d provides the final model in Raudenbush and Bryke's (2002) multiple equation form.

Primary analyses examining hormonal predictors and interactions. To assess moderation effects on hormonal influences, we performed preregistered multilevel regression analyses on daily surveys, nested within participant, with Extra-Pair Sexual Interest as the outcome variable and fixed effects identical to those listed above, with Estradiol level and Progesterone level (both log-transformed) substituted for Conception Probability.

We once again, used Matuschek et al.'s (2017) iterative procedures to model random slope variation, across participants, for within-woman Estradiol and Progesterone. See SOM S1d. For all models, random intercepts and slope terms for both Estradiol and Progesterone were included.

SOM S1d provides the final model in Raudenbush and Bryke's (2002) multiple equation form.

Robustness analyses. We report our primary preregistered analyses in text. But we ran additional analyses using simplifications (e.g., ones dropping Study effects, as few interactions emerged, and between-woman hormone effects, which are orthogonal to effects of interest) and alternative treatments (e.g., ones using raw rather than log-transformed hormone levels). We also selected on one variable we did not use as an inclusion criterion but the BioCycle did (BMI between 18 and 35). And we ran analyses on Study 1 and Study 2 separately. We report these results in SOM, S3.

Follow-up analyses on components. Primary preregistered analyses examined effects on an overall composite measure of extra-pair sexual interests. But we followed up with analyses examining

effects on its two components: Extra-Pair Sexual Attraction and Interest in Extra-Pair Sex.

Forest plots. To summarize moderation effects of interest, we created forest plots of these effects (with 95% confidence intervals), as estimated within our primary preregistered, simplified, and alternative analyses. For additional details, see SOM S1d.

Simple effects analyses. To interpret significant Partner Sexual Attractiveness moderation effects on influences of Conception Probability, Estradiol, or Progesterone, we separately examined Conception Probability, Estradiol, or Progesterone simple effects when Partner Sexual Attractiveness was 1 *SD* above the mean and 1 *SD* below the mean (i.e., effects when women judged partners to be relatively sexually attractive and relatively sexually unattractive). To do so, we zero-centered $\bar{x}$ -scored Partner Sexual Attractiveness at 1 or -1. We estimated these effects utilizing our full preregistered model, though, as we discuss below, moderation effects are similar in simplified and alternative analyses.

Exploratory analyses on In-Pair Sexual Interest. We did not preregister predicted moderation effects on In-Pair Sexual Interests, as neither Arslan et al.'s (2021) corrected analyses (Gangestad & Dinh, 2021) nor Dinh et al.'s (2022b) study found them. But for completeness, we explored effects on In-Pair Sexual Interest and its components, In-Pair Sexual Attraction and Initiation of Sex.

Exploratory analyses on additional measures of sexual interest. Similarly, though we did not preregister predicted effects on General Sexual Desire, Partner-initiation of Sex, and Rejection of Partner-initiated Sex, for completeness we performed analyses on these outcome variables.

2.4 Transparency and openness

Both studies and sets of analyses were fully preregistered on OSF. Data are fully shared, as are R scripts used to manage and analyze data and all output files. See [OSF](#).

3. Results

3.1 Sample characteristics

Table 1 presents descriptive statistics. Overall, mean age was 29.9 ($SD = 3.8$; range = 19 to 36), mean relationship length 8.1 years ($SD = 4.4$; range = .25 to 20.6). 83.9% of women lived with partners; 43.4% were mothers. Study 1 participants were slightly older, had longer relationships, and more often lived with partners, but differences are modest. Both samples were older, had longer relationships, and more often had children compared to previous studies. (In Arslan et al., 2021a/Dinh et al., 2022b, mean age was 25.5/20.9 years, mean relationship length 3.8/2.1 years, 12%/5% were mothers. Earlier studies [Pillsworth & Haselton, 2006; Haselton & Gangestad, 2006; Larson et al., 2012; Shimoda et al., 2018] recruited largely college samples, mean ages 19.5-21.4.)

Table 1

Descriptive statistics by study

Variable	Study 1	Study 2	<i>test(df)</i>	Statistic	<i>p</i>
Age	30.3 (3.6)	29.4 (4.0)	<i>t</i> (468)	2.65	.008
Relationship Length (years)	8.5 (4.3)	7.5 (4.6)	<i>t</i> (472)	2.48	.013
Cohabiting with Partner	88.8%	78.5%	χ^2	3.08	.002
Has Child	47.4%	39.1%	χ^2	1.85	.064
Has Child with Current Partner	45.8%	37.3%	χ^2	1.89	.059

Note. Age and relationship length values are *M (SD)*.

3.2 Primary analyses examining moderation effects of Conception Probability and hormonal associations with Extra-Pair Sexual Interest

Table 2 presents preregistered analyses examining moderation of the association of Conception Probability with Extra-Pair Sexual Interest by Partner Sexual Attractiveness. Robust moderation emerged, $\gamma = -.039$, $t(424.6) = -3.74$, $p = .0002$. When partners were less sexually attractive, conception probability, as it varied within women, more positively predicted extra-pair sexual interests. (For complete set of effects, see SOM, S3.1)

Table 2

Key associations from preregistered fixed effects models with Probability of Conception and Estradiol and Progesterone

Terms	Estimate	CI	<i>t</i>	<i>df</i>	<i>p</i>
Probability of Conception Model					

Study	0.02 [-0.05, 0.09]	0.55	472.1	.583
Probability of Conception (WW)	0.06 [0.04, 0.08]	5.85	447.1	<.001
Partner Sexual Attractiveness	-0.15 [-0.22, -0.08]	-4.17	473.3	<.001
Study * Probability of Conception (WW)	0.00 [-0.02, 0.02]	0.28	428.0	.783
Study * Partner Sexual Attractiveness	0.03 [-0.04, 0.10]	0.81	478.7	.417
Probability of Conception (WW) * Partner Sexual Attractiveness	-0.04 [-0.06, -0.02]	-3.74	424.6	<.001
Study * Probability of Conception (WW) * Partner Sexual Attractiveness	0.00 [-0.02, 0.02]	0.38	446.6	.705
Estradiol and Progesterone Model				
Study	0.01 [-0.05, 0.08]	0.41	467.6	.685
Estradiol (WW)	0.05 [0.02, 0.07]	3.57	529.1	<.001
Progesterone (WW)	-0.08 [-0.10, -0.05]	-6.07	433.6	<.001
Partner Sexual Attractiveness	-0.16 [-0.23, -0.09]	-4.42	477.8	<.001
Study * Progesterone (WW)	0.00 [-0.03, 0.02]	-0.26	433.8	.794
Study * Estradiol (WW)	0.00 [-0.02, 0.02]	0.00	428.2	.998
Study * Partner Sexual Attractiveness	0.01 [-0.06, 0.09]	0.40	470.4	.688
Estradiol (WW) * Partner Sexual Attractiveness	-0.03 [-0.06, -0.00]	-2.31	538.7	.021
Progesterone (WW) * Partner Sexual Attractiveness	0.05 [0.02, 0.08]	3.91	429.2	<.001
Study * Estradiol (WW) * Partner Sexual Attractiveness	0.02 [-0.01, 0.04]	1.50	432.6	.135
Study * Progesterone (WW) * Partner Sexual Attractiveness	0.00 [-0.02, 0.03]	0.04	429.9	.969

Notes. WW = within-woman. Results for the primary interaction effects of interest are bolded. Statistics reported in the table are from the full preregistered models. For reporting purposes, we only included the key associations; terms for menses, self sexual attractiveness, and between-women probability of conception, estradiol, and progesterone, and any interactions associated with those terms were included in the models but are not displayed here. See SOM tables 3.1.1 and 3.1.2 for full model tables.

Table 2 also presents results of primary preregistered analyses examining moderation of hormonal associations. Partner Sexual Attractiveness moderated the associations of both within-woman Estradiol and Progesterone with Extra-Pair Sexual Interest, $\gamma = -.031$, $t(538.7) = -2.31$, $p = .021$, and $\gamma = .049$, $t(429.2) = 3.91$, $p = .0001$, respectively. When partners were less sexually attractive, estradiol levels, as they varied within women, more positively predicted extra-pair sexual

interests and progesterone levels, as they varied within women, more negatively predicted extra-pair sexual interests. (For complete set of effects, see SOM, S3.1)

3.3. Robustness analyses

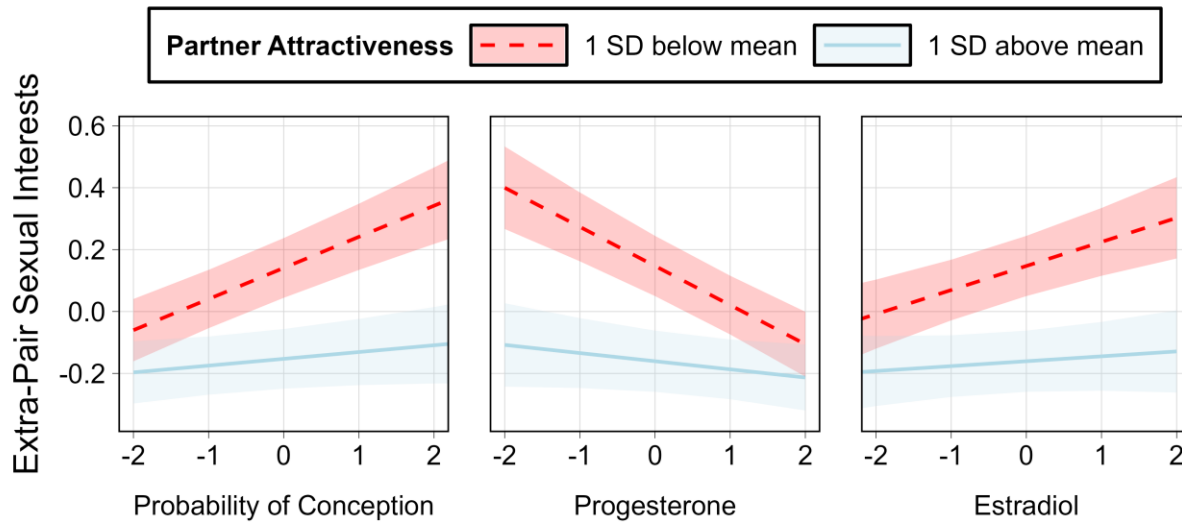
Simplified models (e.g., removing Study effects, between-woman conceptive status/hormone effects, or self-attractiveness effects) and alternative models (e.g., using raw hormone levels) yielded very similar results. Separate analyses on Study 1 and Study 2 yielded significant moderation effects for conceptive status and progesterone for both studies; moderation of estradiol associations was significant in Study 1 but not Study 2. When we selected the subsample who met the BioCycle's criterion for inclusion based on BMI (between 18 and 35), moderation effects for conceptive status, estradiol, and progesterone associations increased by 35%-37%, a pattern consistent with this criterion excluding some participants for whom imputations based on the BioCycle study are less valid. See SOM S3 for a forest plot of effects in these analyses.

3.4 Simple effects

Simple effects analyses were conducted on full models with all controls to interpret moderation effects. When partners were relatively sexually unattractive (1 *SD* below the mean), within-woman Conception Probability positively predicted women's Extra-Pair Sexual Interest, $\gamma = .100$, 95% CI [0.07, 0.13], $t(437.4) = 6.73$, $p < .00001$. By contrast, when partners were relatively sexually attractive (1 *SD* above the mean), within-woman Conception Probability did not significantly predict women's Extra-Pair Sexual Interest, $\gamma = .022$, 95% CI [-0.01, 0.05], $t(437.3) = 1.51$, $p = .131$. See Figure 1 for details.

Figure 1

Simple effects of probability of conception, progesterone, and estradiol on extra-pair interests for women with unattractive and attractive partners



Note. Partner sexual attractiveness was roughly normally distributed ($skew = -0.12$); this resulted in 13.2% of women being at least 1 *SD* below the mean and 20.7% of women being at least 1 *SD* above the mean. Simple effects were estimated via post-hoc tests of the interaction effects using the *emmeans* R package (Lenth & Piaskowski, 2025).

Similarly, when partners were relatively sexually unattractive, within-woman Estradiol levels positively and within-woman Progesterone levels negatively predicted women's Extra-Pair Sexual Interest, $\gamma = .076$, 95% *CI* [0.04, 0.11], $t(531.3) = 4.06$, $p = .00006$ and $\gamma = -.127$, 95% *CI* [-0.16, -.09], $t(432.3) = -7.02$, $p < .00001$, respectively. But when partners were relatively sexually attractive, neither within-woman Estradiol levels nor within-woman Progesterone levels significantly predicted women's Extra-Pair Sexual Interest, $\gamma = .017$, 95% *CI* [-0.02, 0.05], $t(540.9) = .39$, $p = .362$ and $\gamma = -.027$, 95% *CI* [-0.06, 0.05], $t(433.42) = -1.48$, $p = .139$, respectively.

We estimated the effect size (Cohen's *d*) of conceptive status (1 = conceptive, 0 =non-conceptive) on within-woman Extra-Pair Sexual Interest when partners were relatively sexually unattractive and partners were relatively sexually attractive, conservatively disattenuated for imperfect measurement. When partners were relatively sexually unattractive, Cohen's *d* for fertility status was estimated to be .61, a medium-large effect (95% *CI*: .43-.80). When partners were

relatively sexually attractive, Cohen's d was .11, a negligible effect (-.06-.28). See SOM S4 for calculation details and estimated d , not attenuated for imperfect measurement.

3.5 Follow-up analyses on components of Extra-Pair Sexual Interest

Separate analyses on Extra-Pair Sexual Attraction yielded a highly significant within-woman Conception Probability $\times$ Partner Sexual Attractiveness interaction effect, $\gamma = -.052$, 95% CI [-0.08, -0.03], $t(419.4) = -4.42$, $p = .00001$, as well as highly significant within-woman Estradiol $\times$ Partner Sexual Attractiveness and within-woman Progesterone $\times$ Partner Sexual Attractiveness interactions, $\gamma = -.053$, 95% CI [-0.08, -0.03], $t(426.9) = -3.77$, $p = .0002$, and $\gamma = .061$, 95% CI [0.03, 0.09], $t(428.8) = 4.31$, $p = .00002$, respectively. See SOM S5 for details.

Moderation effects on Interest in Extra-Pair Sex ran in the same direction but, aside from the interaction with progesterone levels, fell short of statistical significance: Conception Probability $\times$ Partner Sexual Attractiveness: $\gamma = -.015$, 95% CI [-0.03, -0.01], $t(441.4) = -1.54$, $p = .125$; within-woman Estradiol $\times$ Partner Sexual Attractiveness, $\gamma = -.014$, 95% CI [-0.04, 0.01], $t(452.0) = -1.16$, $p = .247$; within-woman Progesterone $\times$ Partner Sexual Attractiveness: $\gamma = .023$, 95% CI [0.00, 0.04], $t(431.9) = 2.07$, $p = .039$. See SOM S6.

For simple effects of conceptive status on Extra-Pair Sexual Attraction, estimated Cohen's d for women with relatively unattractive and attractive partners (1 SD below and above the mean) were slightly stronger than simple effects on Extra-Pair Sexual Interest: .69 (CI : .51-.88) and .10 (CI : -.07-.26), respectively. See SOM S4.

3.6 Exploratory analyses examining moderation effects on In-Pair Sexual Interest

Based on past work, our preregistration focused on partner sexual attractiveness moderation of associations with extra-pair sexual interests, but we also examined moderation effects on in-pair sexual interests. We report analyses in SOM S7.1.

The Conception Probability $\times$ Partner Sexual Attractiveness effect on the composite measure of In-Pair Sexual Interests was positive but fell short of conventional statistical significance, $\gamma = .015$, 95% *CI* [-0.00, 0.03], $t(439.8) = 1.66$, $p = .097$. Separate analyses on the two components of In-Pair Sexual Interests revealed different patterns. The moderation effect on In-Pair Sexual Attraction was minimal, $\gamma = .002$, 95% *CI* [-0.02, 0.02], $t(456.8) = .22$, $p = .825$, whereas moderation on Sexual Initiation was significant, $\gamma = .023$, 95% *CI* [0.00, 0.04], $t(432.3) = 2.50$, $p = .013$. Simple effects showed that women with both attractive (1 *SD* above the mean) and relatively unattractive (1 *SD* below the mean) partners initiated more sex when conceptive, but with effect size more than double for women with attractive partners: $\gamma = .082$, 95% *CI* [0.06, 0.11], $t(460.2) = 6.35$, $p < .00001$ vs. $\gamma = .037$, 95% *CI* [0.01, 0.06], $t(452.1) = 2.84$, $p = .005$.

Analyses substituting hormone levels for conception probability found similar effects (in the opposite direction) for progesterone but not estradiol levels. For details, see SOM S7.1.

3.7 Additional exploratory analyses

In exploratory analyses, we examined Conception Probability $\times$ Partner Sexual Attractiveness interaction effects on several additional variables related to sexual interests: General Sexual Desire, Frequency of Sex, Partner-Initiations of Sex (with number of rejections controlled) and Rejections of Partner-Initiated Sex (with number of partner initiations controlled). As we predicted no effects, we report analyses in SOM S7.2. In short, we detected no moderation effects except for Frequency of Sex, consistent with moderation on women's own initiation of sex.

4. Discussion

Partner sexual attractiveness moderation of the association of naturally cycling women's conceptive status with their extra-pair sexual interests, as they vary across the cycle, has been of considerable empirical and theoretical interest. Past large-scale replication studies (e.g., Arslan et al., 2021a; Gangestad & Dinh, 2021; Dinh et al., 2022b) provide suggestive but not definitive evidence

for moderation. The current preregistered studies offer strong evidence for it, at least within the population we sampled and using measures we implemented. A first study yielded support but, given contention in this area, we preregistered a second study, also supportive, prior to publication. The probability of the combined results if no moderation exists is near-zero, .0002.

These analyses concern a functional state of interest, women's conceptive status. Yet scholars have also been interested in assessing the potential impacts of physiological states that regulate conceptive status. Hence, we conducted further analyses examining moderation of associations of both estimated estradiol and progesterone levels with extra-pair sexual interests. These analyses yielded evidence for moderation of associations with both estimated hormone levels. Consistent with related associations (e.g., on sexual desire; Dinh et al., 2022a), moderation of progesterone associations was especially strong, moderation of estradiol associations less robust. Validities of estimated estradiol levels are lower than those for progesterone (by ~20% lower), which may well explain some of the difference. (Naturally, in light of associations of estradiol and progesterone levels with conceptive status, these analyses are not statistically independent of analyses examining moderation of associations with conceptive status. At the same time, the latter moderation is consistent with moderation of associations with either estradiol or progesterone levels, or both hormone levels. These follow-up analyses, then, non-redundantly address potential influences by physiological states.)

Prior to discussing theoretical implications, we address some issues that pertain to the empirical phenomenon studied. As philosophers have repeatedly observed, empirical studies do not test isolated, single propositions (e.g., Lakatos, 1970). Their specific procedures invoke a host of background conditions and auxiliary assumptions (e.g., concerning measurement) aside from key propositions of interest. If observations do not fit expectation, one does not know which propositions or assumptions are responsible, a concept of *holist underdetermination* (Stanford, 2023).

And when a study *does* offer support, it does so within a web of conditions and auxiliary assumptions, best explicitly recognized.

Our studies utilized specific measures of predictors, outcomes, and moderator. We estimated conception probability and hormones via backward counts from onset of next menses. These estimates have measurement error. At the same time, (a) alternative means of estimating hormone levels (salivary or urinary assays) have as much or more error; (b) more valid measures (e.g., via assays of serum samples) reduce noise, with effects expected to be *larger*, not smaller; here, errors shouldn't drive positive findings. (Indeed, with women not meeting the BioCycle's inclusion criterion for non-extreme BMI excluded, moderation effects strengthened.) We measured extra-pair sexual interest in particular ways and preregistered two modestly correlated components, extra-pair sexual attraction and interest in extra-pair sex (with a hypothetical attractive target). The moderation effect on the former was larger than that on the latter. Interpretation should attend to this pattern. Finally, women themselves rated partner sexual attractiveness (as well as their own sexual attractiveness). Our items very explicitly asked about *sexual* attractiveness. Not all tests of moderation have done so. Earlier, we discussed Schleifenbaum et al.'s (2024) measure. Not only did it did not explicitly refer to sexual attractiveness; it shows lack of validity for measuring women's perceptions of partner attractiveness. Some measures in other studies (e.g., Dinh et al., 2022b) too did not explicitly ask women to rate partner *sexual* attractiveness. At the same time, we note that moderation by women's ratings of partner sexual attractiveness could be driven by factors other than consensually agreed-upon sexually attractive features, a point we further discuss below.

The standardized interaction effect observed on our overall preregistered measure was $-.039$ ($SE = .010$). We may ask: Is this effect of meaningful size? As the expected effect is an attenuated (vs. a cross-over) interaction (see, e.g., Blake & Gangestad, 2020), we focus on simple effects of conceptive status for women with partners they report to be sexually unattractive (1 *SD* below the

mean) and sexually attractive (1 *SD* above the mean). For women with partners they reported to be relatively sexually unattractive, conceptive status strongly predicted extra-pair sexual interest, effect comparable to $d = .61$ (disattenuated for less-than-perfect reliability/validity of measures), a medium-large effect. No such association was detected for women with partners they deemed to be relatively sexually attractive, $d = .11$. This pattern of moderation appears to be of meaningful size.

As just implied, however, effect size may depend upon conditions and auxiliaries. Indeed, given different validities of measures, true effect sizes almost certainly *do* differ by precise study design and measurement. They may also vary by population. Our sample was somewhat older and, hence, may have had fewer anovulatory cycles than other samples (see, e.g., Hambridge et al., 2013). And we utilized inclusion criteria not used by some others (see SOM S4). Aside from the current studies, Arslan et al.'s (2021a) similar daily survey study offers the most comprehensive published investigation of moderation, though with a different composite measure of extra-pair sexual interest and way of measuring women's perceptions of partner attractiveness. In their corrected data, the interaction effect involving a preregistered measure of partner short-term attractiveness was $-.039$ in their preregistered sample of 143 women and $-.042$ in a robustness sample of 253 women meeting inclusion criteria for our sample—very similar to our own studies ($SE = .015$ -. $.016$; SOM S4 for details⁴). We estimate d to be $.53$ and $.03$, respectively, in their preregistered sample and $.73$ and $.18$ in the selected robustness sample, also highly similar.

Dinh et al.'s (2022b) study examined 180 women's sexual interests four times across the cycle, as opposed to near-daily, thereby offering far fewer total observations and standard errors 2-3

⁴ We acknowledge that estimated effect sizes are imprecise but, for the sake of providing a more nuanced examination of findings above and beyond p -values (e.g., Amrhein et al., 2019), we consider estimation useful. Of Arslan et al.'s full robustness sample of 421 participants, 40% would not have been included in our studies, either because their cycles varied by more than 5 days (84) or they used a contraceptive pill in the past three months (31), were breastfeeding (10), or were 36+ years of age (67). As should be expected, effect sizes estimated from the full, less "clean" robustness sample are somewhat less disparate— $d = .54$ and $.22$ —but don't affect conclusions. See SOM S4.

times those of the studies just discussed. Overall findings were mixed, with moderation sizable for progesterone and modest (and non-significant) for contraceptive status (which, not being the primary focus of the study, was available for only a subset of participants). SOM S4 discusses details.

Nonetheless, differences between effects in that study from the current studies and Arslan et al.'s (much less than a single *SE*) could readily be due to sampling variability, not systematic differences.

Despite almost certain moderation by measurement and, possibly, population, then, observed differences across the studies we discuss above appear to be small to modest, relative to standard errors. We see no *compelling empirical evidence* for systematic variation in effects estimated across these studies, or what features actually do moderate effects.

As noted above, however, *within* our studies, a notable difference in effect size emerged. We preregistered a composite of two components: Extra-Pair Sexual Attraction and Interest in Extra-Pair Sex. Moderation for Extra-Pair Sexual Attraction was highly robust, *p* of 1 in 100,000, standardized effect = -.052, estimated *ds* = .69 and .10. The moderation effect on women's willingness to say that they would have sex with an attractive male (i.e., the one item scenario) was smaller and, in most analyses, not statistically significant. The base rate of stated willingness to have extra-pair sex with an attractive man was very low, more than half of women reporting no interest at all. When partnered with men they perceived as sexually unattractive, then, women in our sample clearly experienced greater attraction to men other than their partners especially when contraceptive (or, more precisely put, *certain* men, the features of which we cannot specify at this point). Yet, in the parlance of Finkel's (2014) I³ model of motivational processes, contraceptive women's "impellance" to be attracted to certain extra-pair men when partnered with a relatively unattractive man was likely "inhibited" and hence very rarely acted upon in this population. Moderation effects on women's extra-pair sexual interest certainly need not imply that, even when women perceive partners to be sexually unattractive, they commonly engage in actual extra-pair sex.

That said, we note, Dinh et al. (2022b) reported the reverse pattern: Stronger progesterone effects on willingness to have extra-pair sex than extra-pair attraction. Once again, current samples were older, in longer relationships, and more likely to have children. Perhaps levels of relationship commitment or related constructs (e.g., “bond strength”; Eastwick, 2009; Eastwick & Finkel, 2012) are key to understanding component-specific patterns. Future research may examine this possibility.

4.1 Partner sexual attractiveness moderation effects on in-pair sexual interests

Past research yields mixed evidence that partner sexual attractiveness moderates the impacts of conception probability on in-pair sexual interests (e.g., Larson et al., 2012). Even after correction, Arslan et al.’s (2021a) large study yielded little evidence for moderation (Gangestad & Dinh, 2021). In our studies, partner sexual attractiveness moderated the impact of conceptive status on women’s initiation of sex with partners, but not (detectably) on attraction to partners: Conceptive status more strongly predicted women’s initiation of sex with highly attractive partners, relative to relatively unattractive partners. Future research may assess this effect’s robustness. Naturally, theory must offer understandings of patterns of moderation on in-pair as well as extra-pair sexual interests.

4.2 Theoretical implications

We now turn to theoretical interpretation. The prediction that partner sexual attractiveness moderates associations of conception probability with extra-pair sexual interests was inspired by the idea that women’s sexual psychology differs across phases of the cycle—specifically, women’s sexual interests should be more attuned to choose the right father for offspring when conceptive (ones especially good investors in offspring or possessing indicators of genetic benefits; see Gangestad et al., 2023). The function of women’s extraordinary level of non-conceptive sex remains incompletely understood. But it cannot lead to conception and, hence, women should be less focused on qualities that render sexual partners good sires when non-conceptive. To the extent that male sexual attractiveness reflects qualities that render males better genetic fathers (or, of greater relevance, did

so ancestrally), it may make sense for women to be more exclusively attracted to sexually attractive partners, vs. extra-pair men, when conceptive.

4.2.1 Contrastive underdetermination and limitations of theoretical interpretation

Do the current findings, then, offer support for the theory that underlies prediction of the moderator effect? No, not in any simple, straightforward way. *Contrastive underdetermination* refers to the fact that evidence for a particular empirical pattern does not determine theoretical interpretation (Stanford, 2023). Multiple theoretical interpretations (i.e., accounts of causal processes) of a given empirical pattern are always possible, even if we cannot presently lay out multiple (even single) adequate theoretical interpretations. In the current context, specific theory predicts moderation. But other theories, even if not currently articulated, may do so as well.

And, indeed, theory laid out above faces challenges by other empirical patterns. It implies that women's preferences for sexually attractive features that may have been indicators of genetic quality ancestrally shift across the cycle, thereby driving the partner sexual attractiveness moderation effect. Yet large replication studies have generally not found substantial preference shifts.

The theoretical implications of the current findings, then, are *not* that they straightforwardly offer support for theory that led to their prediction. Rather, in concert with other findings, they importantly introduce to existing narratives a *conceptually unresolved tension between findings*. Interesting moderation exists and demands explanation. Yet what explains it, if not robust preference shifts? Any adequate theory of how conceptive status, through hormonal influences, affects women's sexual interests *must explain this joint pattern of effects*.

We emphasize the upshot here: Empirical findings can have substantial theoretical implications even when they offer no straightforward support for theory—indeed, when we do not currently understand them. In some cases, findings severely challenge prevailing theory. As well, findings that available theories fail to offer adequate understanding of may become theoretically

crucial linchpins precisely because they inspire the development of new theory to explain them. In our view, the current findings have these kinds of implications.

4.2.2 Implications for motivational priorities theory

We start with a theory challenged by the current findings. Under the narrative that preference shifts and the current moderation effect don't robustly replicate, an alternative view of how women's sexual interests change across the cycle emerged. Sex can yield multiple benefits (e.g., increased intimacy with partners, mate evaluation). But only when women are conceptive can it benefit via impregnation. Motivational priorities theory (e.g., Roney, 2018) argues that selection hence shaped women's "general sexual desire" to increase during their conceptive phase, as influenced by estradiol (promoting desire) and progesterone (suppressing it). Factors other than hormones (e.g., social cues that sex may enhance intimacy or serve to evaluate a mate) elicit sexual interest *independently* of hormones. But hormones don't affect their potency. The only distinctive feature of sexual interest when women are conceptive in their cycles is that there's more of it.

The current findings are inconsistent with motivational priorities theory. Again, that theory asserts that contextual factors affect sexual interests *independent* of hormonal inputs. But here, a contextual factor (having a partner who women deem sexually attractive vs. not) *has* different effects on targeted sexual interests (attraction to certain extra-pair men) depending on hormone levels—a pattern compatible with the idea that hormones affect sexual interests related to specific functions and incompatible with the idea that hormones exclusively affect "general sexual desire." We note that this is true if sexually attractive features of partners drive moderation, but also true if a third variable (e.g., relationship satisfaction, as discussed below) drives the effect. Could motivational

priorities theory be modified to be able to explain moderation? Perhaps—though it is not now clear what an adequate (and generative vs. ad hoc; e.g., Lakatos, 1970) modification would look like.⁵

4.2.3 Explorations within a dual sexuality framework

The dual sexuality framework proposes that women's conceptive and non-conceptive sexuality were shaped to secure different kinds of fitness benefits ancestrally. In contrast to motivational priorities theory, it expects that sexual interests during conceptive and non-conceptive phases differ in ways other than sheer strength; given their different functions, conceptive and non-conceptive sexual interests may be affected by somewhat different factors. The current set of findings fits within this conceptualization. That said, the dual sexuality framework is a broad theoretical approach, not a specific set of theoretical claims (e.g., Gangestad et al., 2023). A challenge, then, is to identify what specific form of dual sexuality (if any) can explain both the current set of findings and broader patterns of hormonal associations with sexual interests.

The idea that women's sexual interests should be more attuned to choose the right father for offspring when conceptive (e.g., ones possessing indicators of genetic benefits) falls within the broader dual sexuality framework. Again, it is challenged by lack of robust, replicated preference shifts. But can this view (or, more specifically, this view in conjunction with auxiliary assumptions) be modified (in a generative, not merely ad hoc, way) to explain findings? We discuss possibilities.

4.2.2.1 Might preference shifts exist for some sexually attractive male features? Researchers have examined multiple male features purportedly related to sexual attractiveness and, potentially, (ancestrally) genetic fitness. But the list may be flawed. The most studied preference shift is that for male facial

⁵ Sexual desire in general (mean sexual interests, no matter the target) robustly increases when women are conceptive (e.g., Roney & Simmons, 2013, 2016; Jones et al., 2018a; Arslan et al., 2021a; Shirazi et al., 2019a,b; Marcinkowska et al., 2022; Pierson et al., 2021; Schleifenbaum et al., 2024; see also earlier reviews by Hill, 1988; Regan, 1996), as expected by motivational priorities theory. But here again, contrastive underdetermination arises: Other theories, including the dual sexuality framework, are also compatible with these increases (Gangestad et al., 2023).

masculinity (e.g., Gildersleeve et al., 2014a; Gangestad et al., 2017). Large-scale replication studies indicate that hormonal influences are likely nil (e.g., Jones et al., 2018b; Marcinkowska et al., 2018). At the same time, facial masculinity now appears to be a poor candidate reflective of genetic fitness. It scales similarly with height in both men and women, not expected of sexually selected ornaments associated with good condition (Zaidi et al., 2019). Moreover, male facial masculinity failed to predict mating success or reproductive success across both Western and traditional societies in meta-analyses (Lidborg et al., 2022). Facial symmetry, another widely studied preference shift, is likely also a poor indicator of condition (e.g., Gangestad & Thornhill, 2003; Gangestad, 2022).

Are any candidate features better? In the same meta-analyses, one male masculine feature *did* predict mating success and, in traditional societies, reproductive success: muscularity/strength (Linborg, 2022). And, unlike male facial masculinity, muscularity strongly predicts male sexual attractiveness—notably, men’s bodily attractiveness (e.g., Sell et al., 2017). Early studies of shifts in preferences for facial masculinity typically yielded significant p -values in the .02-.05 range (Gangestad et al., 2017; see Rohrer, 2018). The few studies examining shifts in preferences for muscularity typically yielded p -values less than .01 (Gangestad et al., 2017). Still, large-scale replication studies have yielded mixed results. Stern et al. (2021) found that progesterone levels, as they vary within-woman, predict women’s preferences for bodily strength, the standardized effect close to progesterone effects on sexual desire (Dinh et al., 2022a). Jünger et al.’s (2018a) data yielded an association of progesterone levels with preferences for a near-identical measure of strength for partnered but not unpartnered women (Gangestad et al., 2019a,b; see also reply by Stern et al., 2019). The true preference shift for male strength may be non-zero for partnered women. At the same time, it is likely small—perhaps too small, by itself—to “save” the view that preferences for specific sexually attractive features meaningfully shift.

Lidborg et al. (2022) found mixed evidence that vocal masculinity predicts mating and reproductive success. Replication studies examining shifts in vocal masculinity preferences too show mixed results (e.g., Pisanski et al., 2014; Jones et al., 2018c; Jünger et al., 2018b); preference shifts, if non-zero, too are likely small. But are shifts for other sexually attractive features stronger? As well, might shifts for any one preference be smaller than effects for aggregate preferences contributing to overall sexual attractiveness? Additional research may address these issues.

4.2.2.2 Are preference shifts themselves moderated? Another tack is possible: Perhaps preference shifts are themselves moderated. Women whose partners are sexually attractive show relatively little extra-pair sexual interest, relative to in-pair interest, throughout the cycle but especially when conceptive. The same could be true of women in other conditions (e.g., strongly attached relationships; Eastwick, 2009; Eastwick & Finkel, 2012). It may make little sense for these women to show shifts in preferences for male features, particularly in extra-pair men; after all, they show no marked interest in non-partners, relative to partners, when conceptive. Perhaps only women who, given their conditions, *do* experience extra-pair (relative to in-pair) sexual interest also experience shifts in preferences for male features thought to be indicators of genetic benefits (e.g., strength). If so, preference shifts across *all* women will be non-zero but very modest in size—in line with observed shifts in preferences for muscularity (e.g., Dinh et al., 2022a). Yet *some* women may experience *meaningful* preference shifts. This possibility should be empirically assessed.

4.2.2.3. Might preference shifts not be about sexually attractive features per se? Perhaps partner sexual attractiveness partly reflects other qualities of men or women's relationships with them, which drive the moderation effects documented here. Again, in our studies, and most others examining moderation effects, women themselves—not independent third parties—rated partners' sexual attractiveness. Possibly, factors other than men's consensually agreed-upon sexually attractive features drive moderation. Perhaps, for instance, women more satisfied or bonded with partners rate

them as more sexually attractive, and these factors moderate the association of conceptive status with extra-pair sexual interest. This possibility faces challenges. Conceptually, what specific quality does moderate associations and, crucially, from a theoretical perspective (e.g., within a specific, articulated dual sexuality view), *why* does it do so? E.g., do women especially value bondedness with partners when conceptive (e.g., Eastwick, 2009)? Empirically, available evidence casts doubt on valued relationship qualities doing the “work” of moderation. Arslan et al. (2021a) asked women to rate partners’ attractiveness as a long-term partner as well. With both partner sexual attractiveness moderation and partner long-term attractiveness moderation entered as predictors, the former strengthened; the latter effect descriptively ran in the opposite direction (see Gangestad & Dinh, 2021), a result unlikely if satisfaction with relationship partners or bond strength drives effects. That said, once again, the possibility that some third variable drives moderation should be explored.

4.2.3 Explorations more generally

Again, we emphasize: A tension between empirical findings demands resolution. A broad dual sexuality approach may be a potentially fruitful path forward. That said, identification of *any* alternative theoretical conceptualization that can explain both current and past findings would be a welcome contribution. Again, no understanding of how hormones affect women’s sexual interests is adequate if it doesn’t resolve this tension.

4.6 Summary

In seminal studies, Haselton and her colleagues found that associations between naturally cycling women’s conceptive status and their extra-pair sexual interests are moderated by partners’ sexual attractiveness. Arslan et al. (2021a) claimed that their large-scale replication study found no evidence for this moderation effect. Yet their analyses were fraught with well-documented coding errors. With errors corrected, Arslan et al.’s data were compatible with meaningful moderation. At the same time, additional work to fully assess the robustness of moderation has been needed.

Two large-scale replication studies leave little doubt: True moderation very likely exists, at least within the population we sampled and using measures we implemented. Moderation by partner sexual attractiveness demands explanation. If current theory cannot adequately explain it, novel theory development is needed.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

No AI was used in the writing of this manuscript. *ChatGPT* (*OpenAI*) was used by authors while writing and interpreting some R code, primarily to troubleshoot coding errors. All code used was reviewed by authors. We take full responsibility for the content of the publication.

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